

JINGTIAN WU

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EDUCATION

Cornell University

B.A. in Computer Science, Minor in Artificial Intelligence
GPA: 3.97

May 2026 (expected)

RESEARCH INTERESTS

Natural Language Processing, Multimodality, Machine Learning, Reinforcement Learning

PUBLICATIONS

- [1] **Jingtian Wu** and Y. Hicke, *Joint Training Optimization: Calibrating Paired LLM Personas via Self-Play*, Accepted (Poster), PersonaLLM: Workshop on LLM Persona Modeling @ NeurIPS 2025.
- [2] **Jingtian Wu**, J. Guo, G. Guo, and Q. Yang, “*The AI is Organic*”: *Fostering Music Ideation with Multimodal AI Inspirations*, Under review (Invited for Revise & Resubmit), CHI 2026.
- [3] **Jingtian Wu** and C. Cardie, *Reasoning court: Combining reasoning, action, and judgment for multi-hop reasoning*, 2025. arXiv: 2504.09781 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2504.09781>.
- [4] **Jingtian Wu**, “Algorithmic composition of music utilizing the digits of pi,” in *Highlights in Science, Engineering and Technology*, vol. 85, 2024, pp. 570–577.
- [5] X. Zhao, Y. Zhou, **Jingtian Wu**, Q. Xu, and Y. Zhang, “Turn real people into anime cartoonization,” in *Proc. IEEE ICCECE*, 2021, pp. 387–393. DOI: 10.1109/ICCECE51280.2021.9342433.

RESEARCH EXPERIENCE

LLM Persona Modeling: Behavioral Alignment in AI-Simulated Personas Summer 2025 – Present
Cornell NLP Group | Advisor: Claire Cardie; Collaborator: Yann Hicke Cornell University

- Proposed a **self-play reinforcement learning** framework that jointly trains paired LLM personas to co-adapt and align their behaviors toward a shared target [1].
- Evaluated the method in AI-simulated doctor-patient dialogues, demonstrating data-free behavioral calibration and laying the groundwork for human-like persona simulations in general education and training contexts.
- Currently extending the framework to multi-turn interactions and domain-agnostic settings.

OmniWizz – A Multimodal Music Ideation Assistant Summer 2025 – Present
Advisors: Qi Yang and Ge Guo Cornell University

- Co-led the design and development of OmniWizz [2], an interactive **multimodal system** generating coordinated text, image, and audio outputs to inspire music creation.
- Independently built the full-stack architecture enabling flexible cross-modal input–output workflows that support diverse and innovative creative exploration.
- Conducted a 16-participant study combining quantitative and qualitative analyses, showing OmniWizz expanded ideation, reduced fixation, and preserved creators’ sense of ownership. [View Demo]

Automatic Synthetic Data Generation Spring 2025
Cornell NLP Group | Advisor: Claire Cardie Cornell University

- Exploring automated dataset construction for **self-evolving models**, developing a teacher–student LLM framework inspired by model distillation and curriculum learning, where an expert LLM acts as an in-environment teacher to generate domain-specific questions, decompose tasks, and refine the student model through guided interaction for scalable continual adaptation in medicine.
- Built a hierarchical knowledge map expansion pipeline, covering 226 subfields, generating 2.9K+ teacher–student dialogues and MCQA datasets without human annotation.
- Fine-tuned the student LLM, achieving +6.76% accuracy over baseline with 6.4× fewer training samples.

Culturally and Temporally Contextualized Lyric Generation

Advisor: Matthew Wilkens

Spring 2025
Cornell University

- Investigating how to adapt LLMs for lyric generation that is culturally and temporally grounded.
- Curated a 31,000-song corpus and fine-tuned LLaMA 3.1 8B with LoRA for context-aware generation.
- The resulting model achieved +3.4% CoSENT, +67.4% ROUGE-2, and +1.6% BERTScore improvements, with human evaluators rating its cultural and temporal relevance 36% higher on average.

Automatic Code Generation System

Cornell NLP Group | Advisors: Claire Cardie and Wenting Zhao

Fall 2024
Cornell University

- Extended Aider with a multi-agent system to explore how LLMs can function more effectively as autonomous software engineers
- Designed components for task planning, debugging, and progress tracking, orchestrated through an automatic iteration loop. Proposed safeguards including a Git-based rollback mechanism and iterative refinement module, preserving best-performing function versions.
- On the Commit-0 parser library, the system boosted accuracy by 15.5 percentage points over baseline.

Enhancing Multi-Hop Reasoning in LLMs

Cornell NLP Group | Advisors: Claire Cardie and Wenting Zhao

Summer 2024
Cornell University

- BURE Researcher; awarded \$7,000 fund for research in Summer 2024.
- Tackled LLM hallucinations in multi-hop QA by developing Reasoning Court [3], extending ReAct-style reasoning–retrieval with two agents and an LLM judge that selects or synthesizes factually grounded answers.
- Outperformed few-shot prompting baselines on HotpotQA, FEVER, and MuSiQue, achieving up to +7.2% EM and +10.5% F1 improvements while reducing LLM calls compared to hybrid baselines.

Computer Music Synthesis and Composition

Advisor: Roger B. Dannenberg

Summer 2023
Carnegie Mellon University

- Explored mapping π 's digits to the minor pentatonic scale, using programming to simulate traditional Chinese instruments for mathematically grounded music.
- Designed an algorithmic method to generate a melody and an automated countermelody.
- Composed an original piece and showcased the creative potential of algorithmic music composition.

Algorithms for Big Data

Advisor: David Woodruff

Summer 2021
Carnegie Mellon University

- Tackled inefficiency of GAN-based cartoonization, aiming for lightweight, high-quality anime-style results suitable for resource-constrained devices.
- Sole developer of *LWAnimeGAN* by combining AnimeGAN and photo2cartoonGAN with architectural optimizations—reduced output channels, depthwise separable convolutions, and removal of HourGlass blocks.
- Reduced parameters by 8.8%, FLOPs by 56.1%, and inference time by 18.6% while preserving visual fidelity.

RESEARCH-RELATED PROJECTS

LLM Framework for Academic Peer Review

Advisor: Thorsten Joachims; Collaborator: Zhaolin Gao

Spring 2024
Cornell University

- Led full-stack development of the demonstration system for the research paper *Reviewer2: Optimizing Review Generation Through Prompt Generation*.
- Contributed to the NSF-supported *Frameworks: arXiv as an Accessible Large-Scale Open Research Platform*.
- Presented the system at BOOM 2024 (Bits On Our Minds), featured in the *Cornell Chronicle*.

AWARDS & HONORS

Cornell Nominee, CRA Outstanding Undergraduate Researcher Award

2026

– Selected as one of 2–4 nominees from the Department of Computer Science.

Dean's List of the College of Arts and Sciences

Fall 2022, Spring 2024

TEACHING EXPERIENCE

CS 3700: Foundations of AI Reasoning and Decision-Making, Cornell

Fall 2024, Fall 2025

– Led weekly office hours and guided students through complex topics (e.g., heuristic search, game-play, and reinforcement learning), provided exam preparation support, and responded to questions on Ed discussion.

SKILLS

Languages	Python, JavaScript, Java, L ^A T _E X, Markdown, HTML/CSS
Machine Learning / NLP	PyTorch, Hugging Face Transformers, LoRA/QLoRA, Unsloth
Software Development	FastAPI, Flask, Node.js, React
Tools	Git, Jupyter Notebook, Visual Studio Code

HOBBIES & INTERESTS

Competitive Chess: Tennessee State Champion; 4th place, U.S. National Blitz Championship; multiple-time Hunan Provincial Champion (China); Team Leader for Hunan Province, National Mind Sports Games (China); represented Cornell in the Inter-Ivy Chess Championship; peak Chess.com ratings of 2974 (bullet) and 2674 (blitz).

Music Production: Compose, arrange, and produce original and cover works across multiple genres; 2M+ total views and 400K+ likes on TikTok and Bilibili; 4th place, Cornell CSSA Singing Competition; former University Growth Specialist at Suno, leading campus ambassador outreach for AI music platform adoption.